Build a rain gauge

IoT · Level 1 · Lesson 3



Today we will

- Explain that rainfall is measured in millimetres of depth.
- Build a rain gauge from a bottle, with a zero line, a millimetre scale and a funnel.
- Choose a good place for the gauge and agree a daily reading routine.

Build it, place it, read it

- An adult cuts off the top
- Mark 0 and a mm scale
- Set it in open ground
- Read daily, empty to 0

Make a bottle rain gauge: Build

1. Warm-up (5 min): Ask: “Did it rain a lot or a little last time?” Let pupils disagree.
2. Teach (7 min): Explain millimetres of rain: the depth on flat ground.
3. Build (15 min): Groups add stones and water just to cover them, mark the 0 line, and tape on a paper scale marked every millimetre from a ruler.
4. Place it (8 min): Walk to the spots you chose.
5. Wrap-up (5 min): Back in class, draw a reading rota and a table on the board: date, time, rain (mm).

Before class, cut the top off each bottle where the sides start to curve, and tape the cut edges. Check that each bottle’s sides are straight. Walk the school grounds and pick two or three open spots to discuss.

Let’s discuss

1. Is your 0 line level?
2. Does every millimetre on your scale match the ruler?
3. Is there anything above the gauge that could block rain or drip water in?

Exit ticket

- One rule for reading the rain gauge, and why it matters.